

Digital Photography

CVAI

20. Februar 2025

FH Zentralschweiz



Overview

- Motivation
- Digital photography pipeline
- Taking pictures: Focal lengths, aperture and shutter speed
- Lenses and distortion
- Imaging sensors
- White Balance
- HDR Imaging
- ...



Slide credits

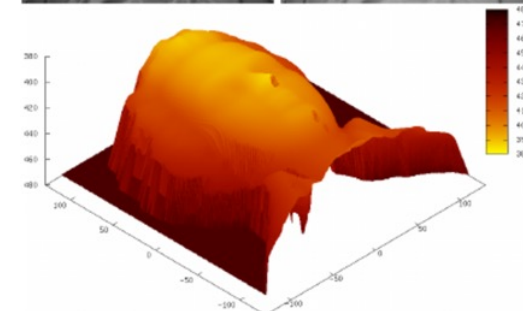
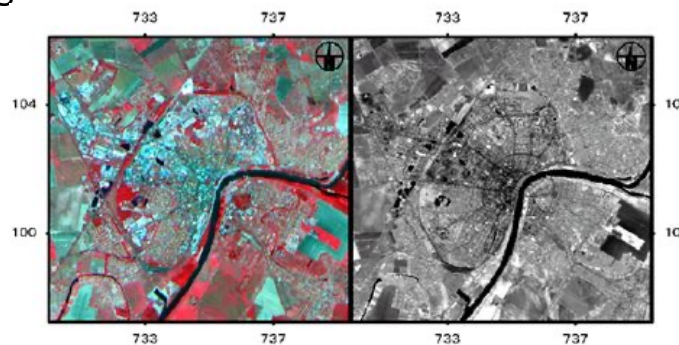
Inspiration and examples from

- CVPR 2016, Understanding the In-Camera Image Processing Pipeline for Computer Vision, Michael S. Brown
- Computer Vision course at CMU (Carnegie Mellon) from Matthew O'Toole
- Computer Vision: Algorithms and Applications, Richard Szeliski
- Images from Adobe Stock Images

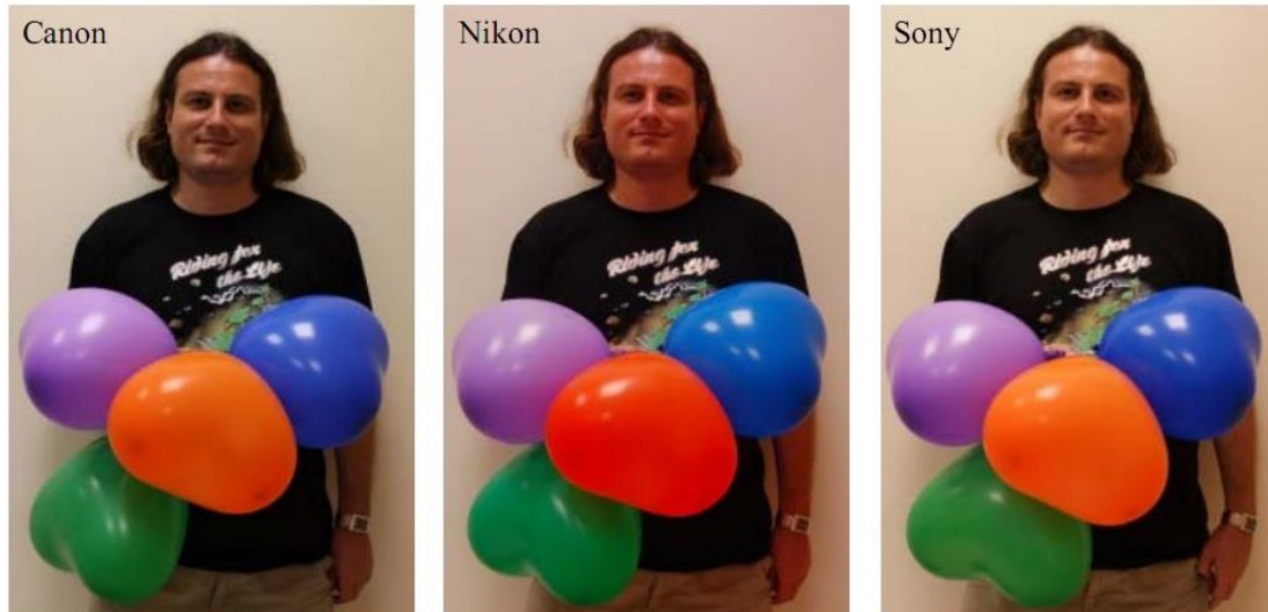
Motivation: Imaging for more than photography

Many application, specially in image processing and computer vision are not interested in the photo, but want to measure or compare the light / radiation.

- Medical imaging
- Satellite images
- Image matching / Multi image processing
- HDR Imaging
- Shape from shading
- ...

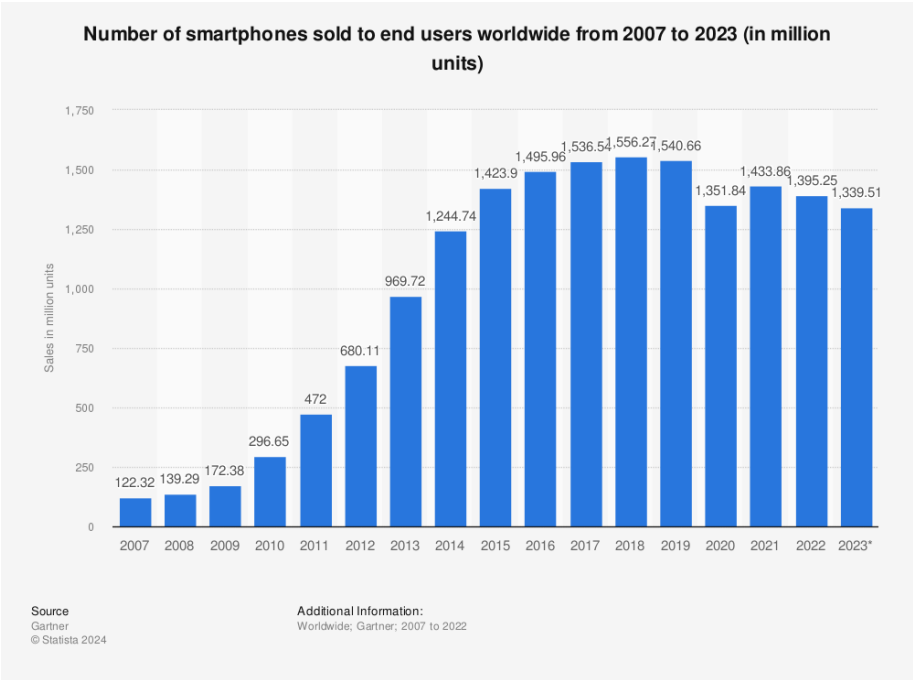


Motivation

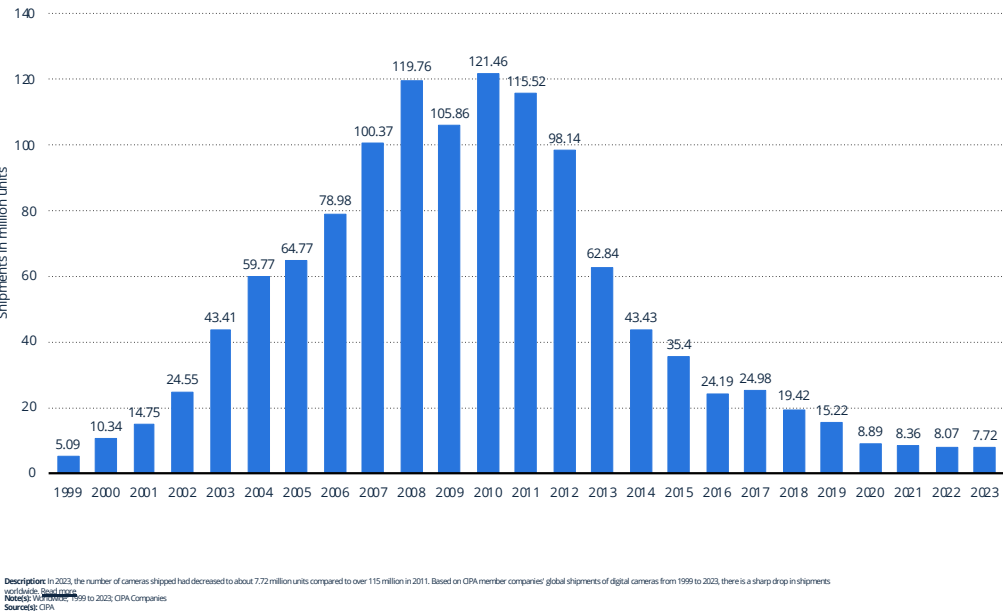


Photos from different cameras, taken with the same settings

Smartphones: the end for cameras?



CIPA companies' shipments of digital still cameras worldwide from 1999 to 2023 (in million units)



Motivation

Q: iPhone 13 ProMax HDR photos oversaturated (i.e look fake)

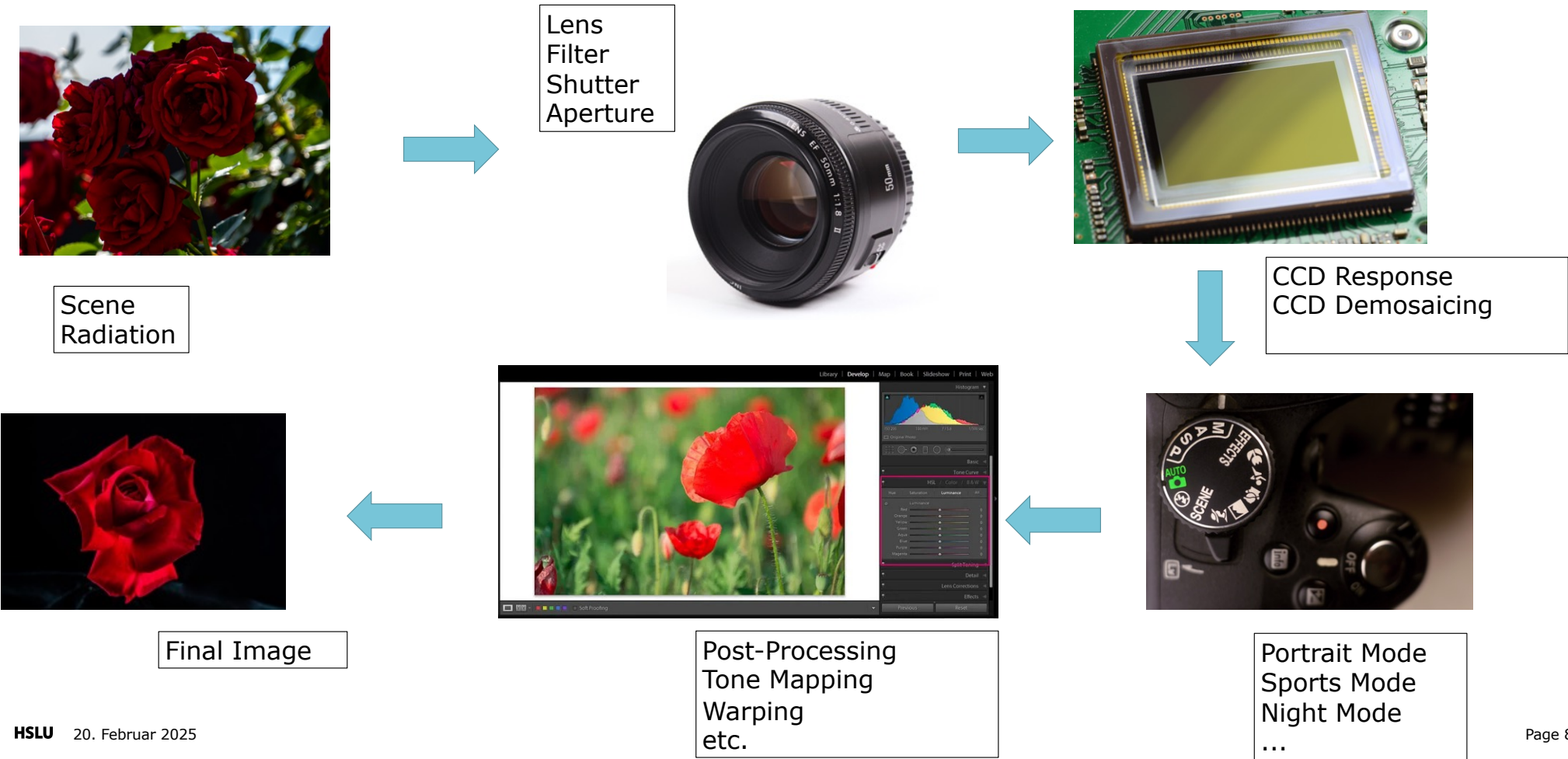
I just got my iPhone 13 Pro Max last night and all my outdoor photos are oversaturated, look fake, and I can't find a way to edit the HDR layers or even turn HDR off. The only solution is to turn down the saturation, but they still look fake, just less saturated.

Am I missing a setting?

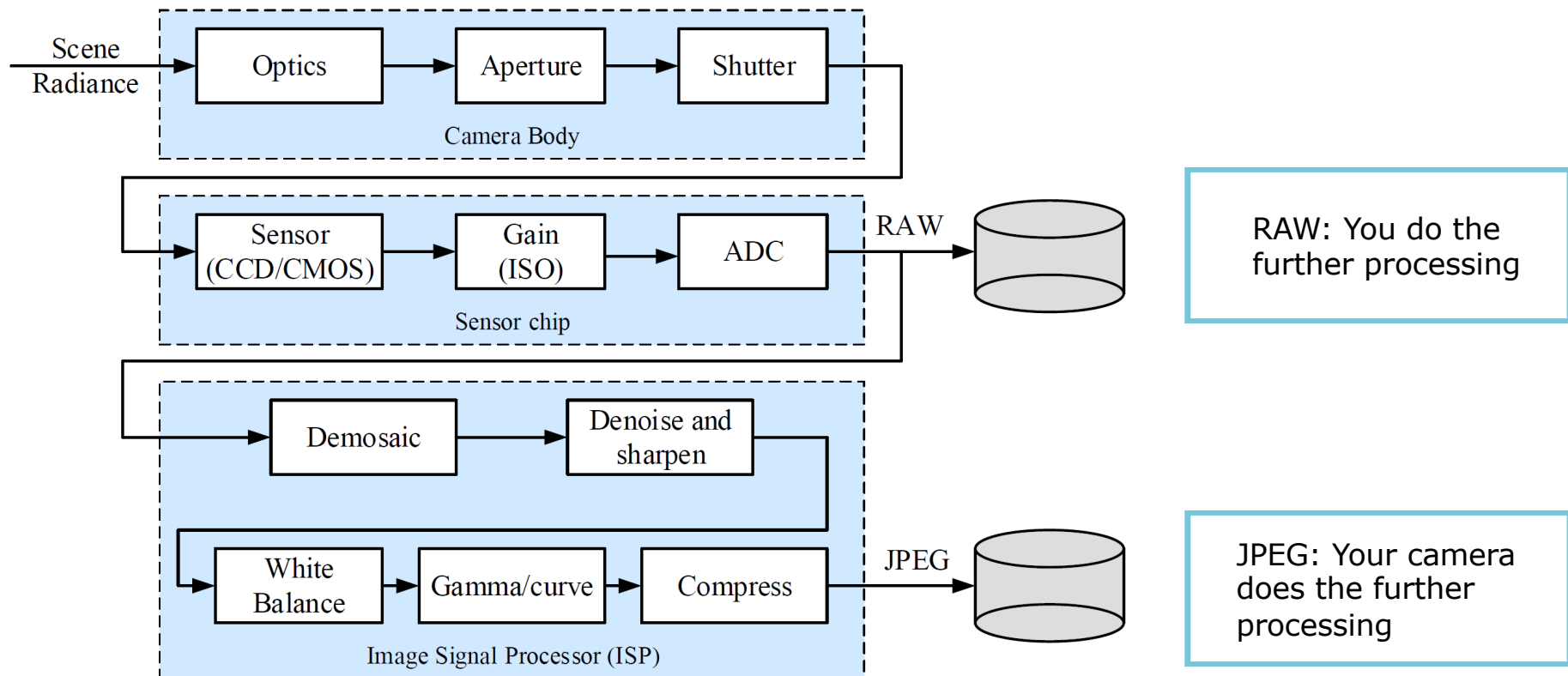


Andrew Hoyle/CNET

Digital photography pipeline

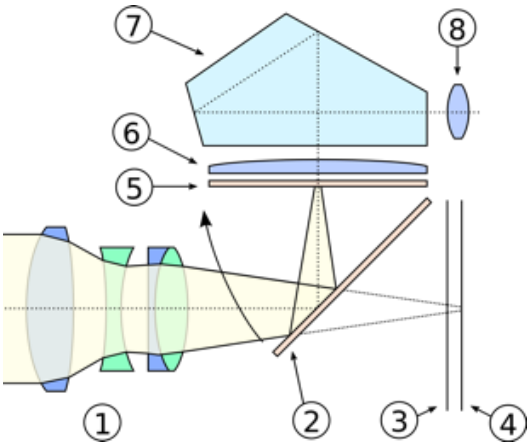
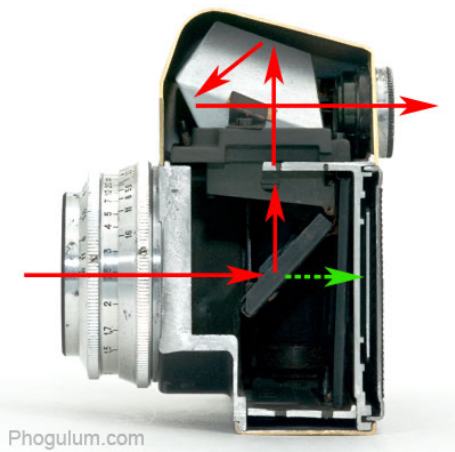
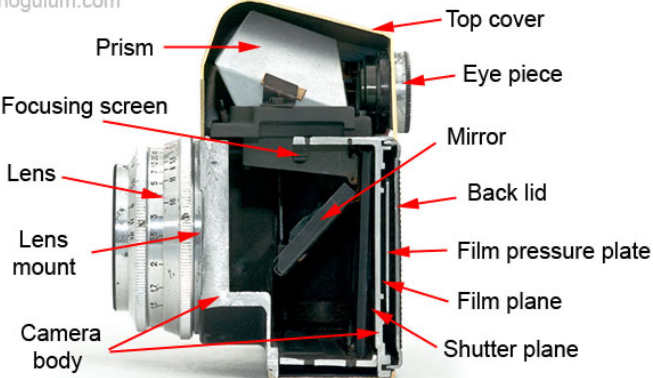


Digital photography pipeline



SLR Cameras

Phogulum.com



Mirrorless cameras



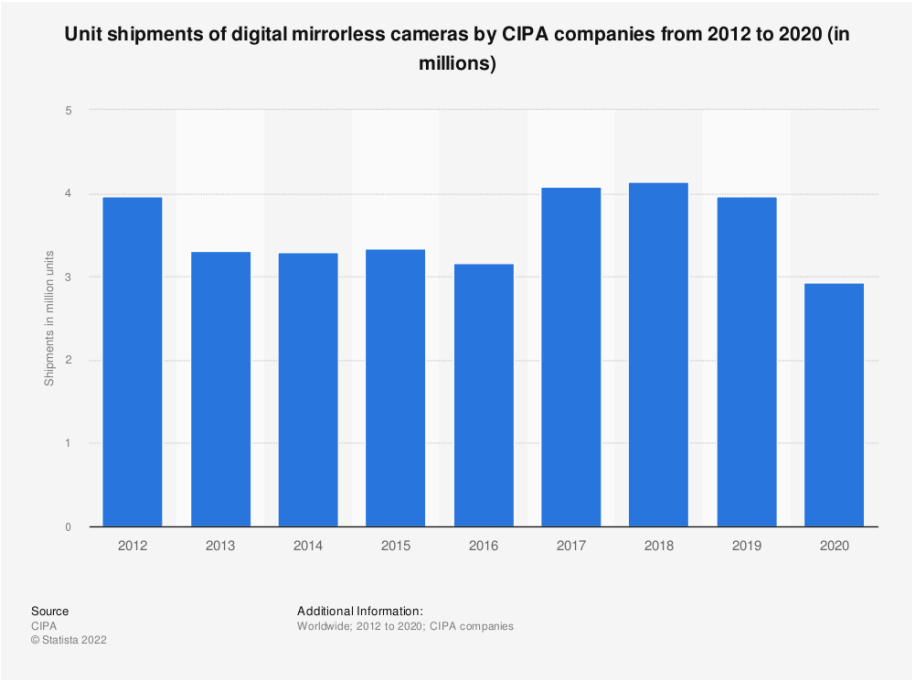
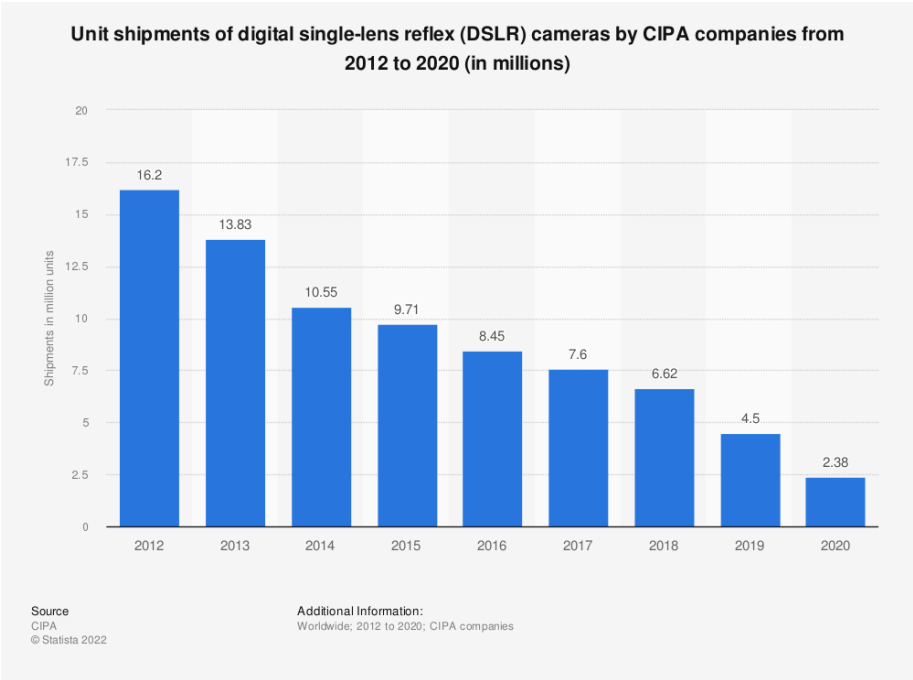
Leica M8 (2006)

- Simpler (no mirror)
- No viewfinder
- Displays image as received by the sensor



Nikon Z5 (2020)

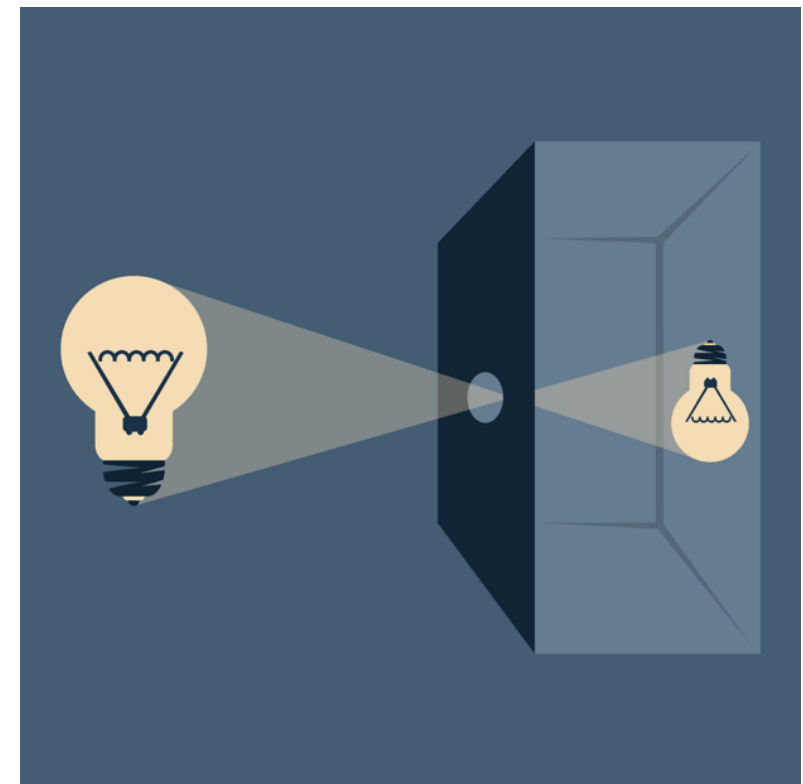
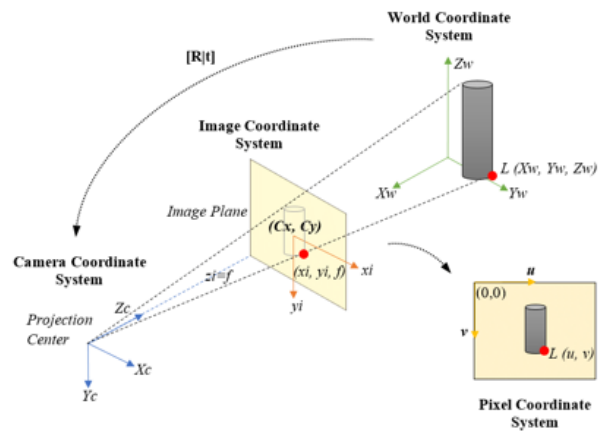
DSLR vs. mirrorless cameras



Pinhole Camera

For most applications, it is fine to treat the camera as an ideal pin-hole camera.

See more in later lectures on 3D reconstruction.

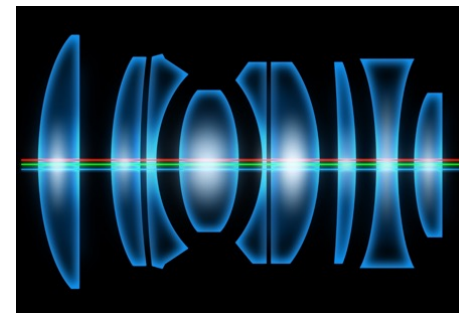
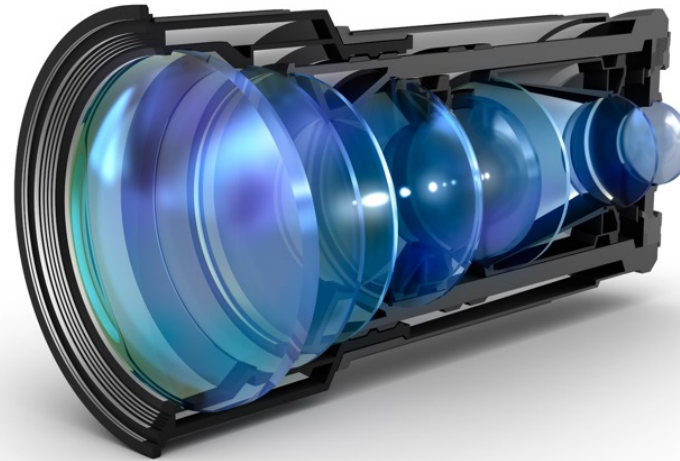


Camera Lenses Basics

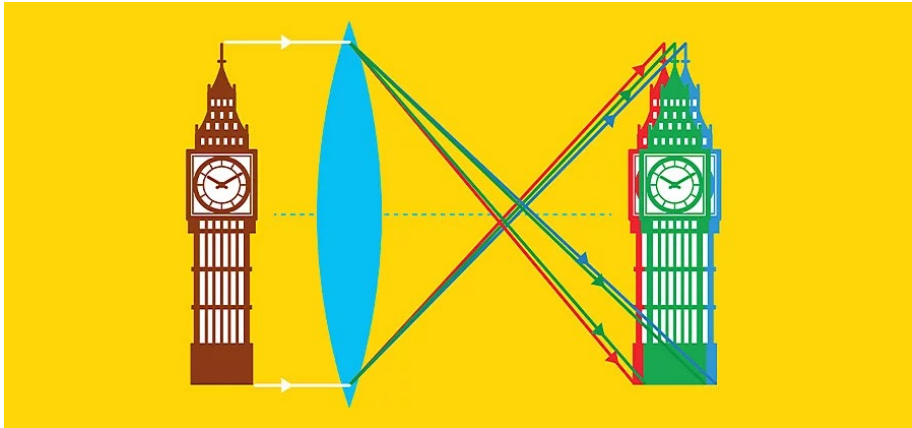
Light passes through (multiple lenses)

Effects:

- Chromatic aberration
- Lens distortion
- Vignetting



Chromatic Aberration

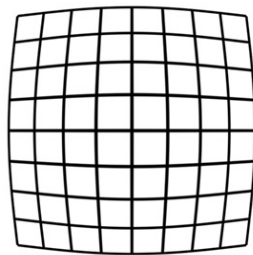


Lenses act like prisms, so light of different wavelengths take a different paths through the lenses

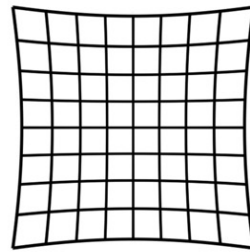
Expensive camera lenses try to minimize this



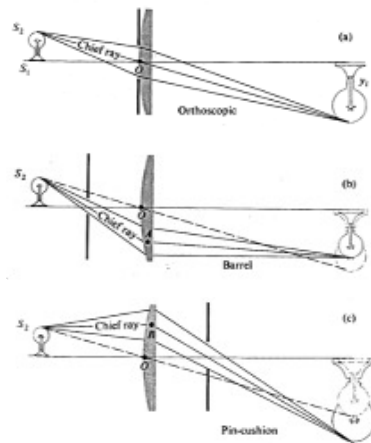
Lens effects and distortion



Barrel Distortion



Pincushion Distortion



Lens distortion

Lens distortions are often modelled by a *radial* model.

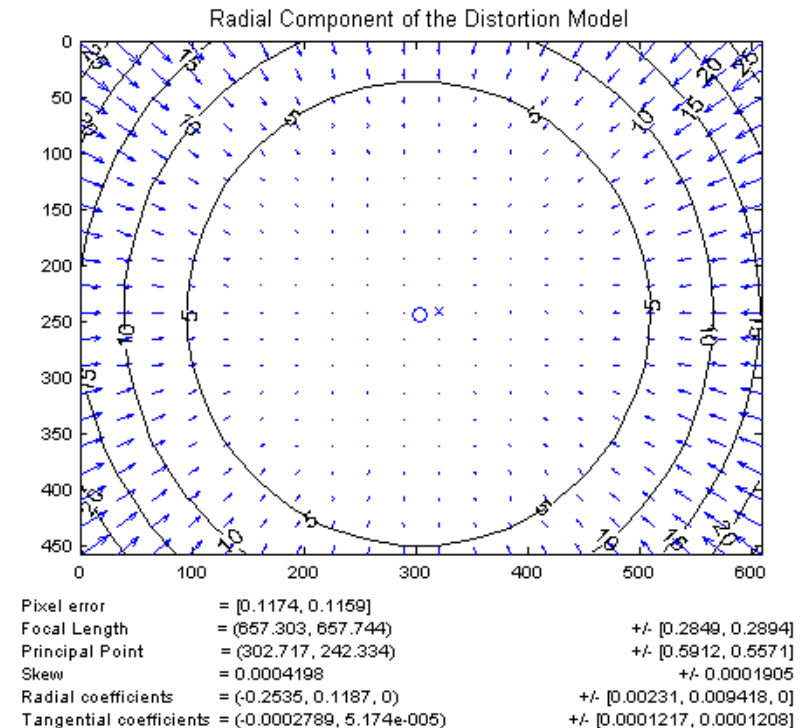
This says that the coordinates in the observed images are displaced in proportion to their radial distance from the center.

If the distortion is towards the center it is called a *barrel* distortion, if it is away a *pincushion* distortion.

A simple model uses low-order polynomials:

$$\hat{x}_c = x(1 + \kappa_1 r_c^2 + \kappa_2 r_c^4)$$

$$\hat{y}_c = y(1 + \kappa_1 r_c^2 + \kappa_2 r_c^4)$$



http://www.vision.caltech.edu/bouquetj/calib_doc/htmls/example.htm

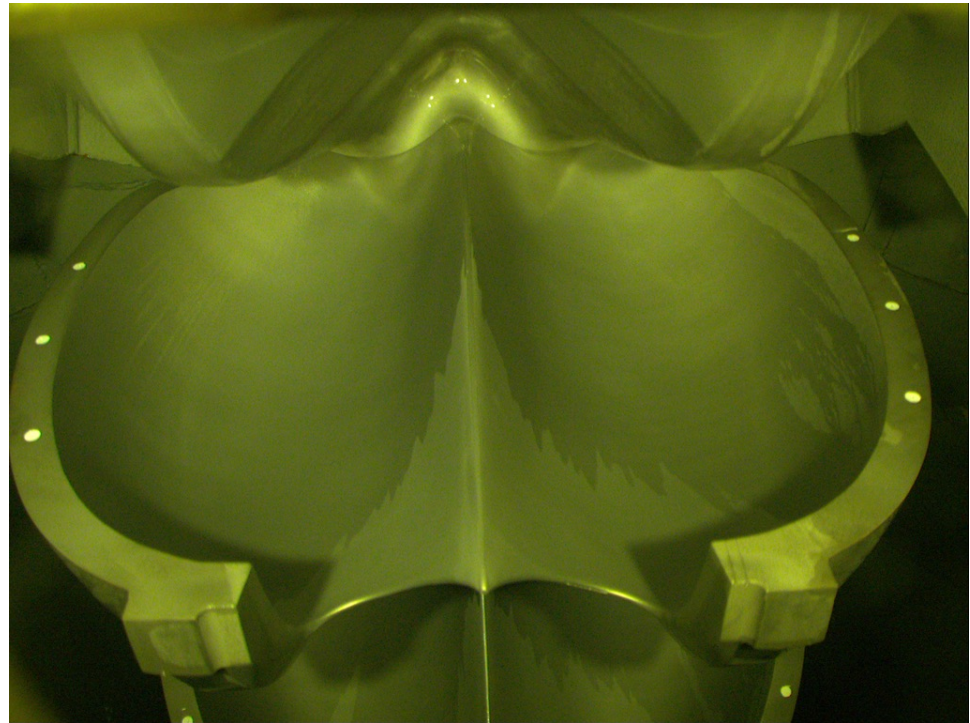
Example from photography



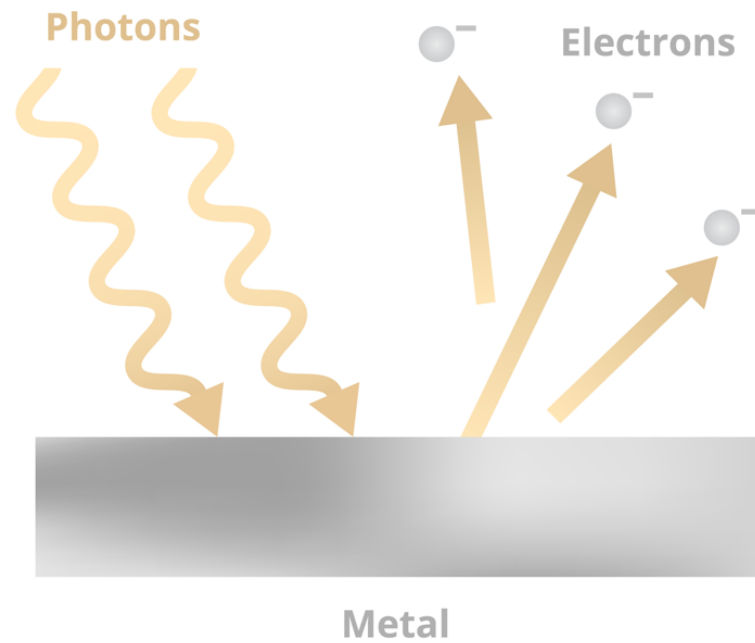
<https://www.panotools.org/dersch/barrel/barrel.html>

Example from image processing

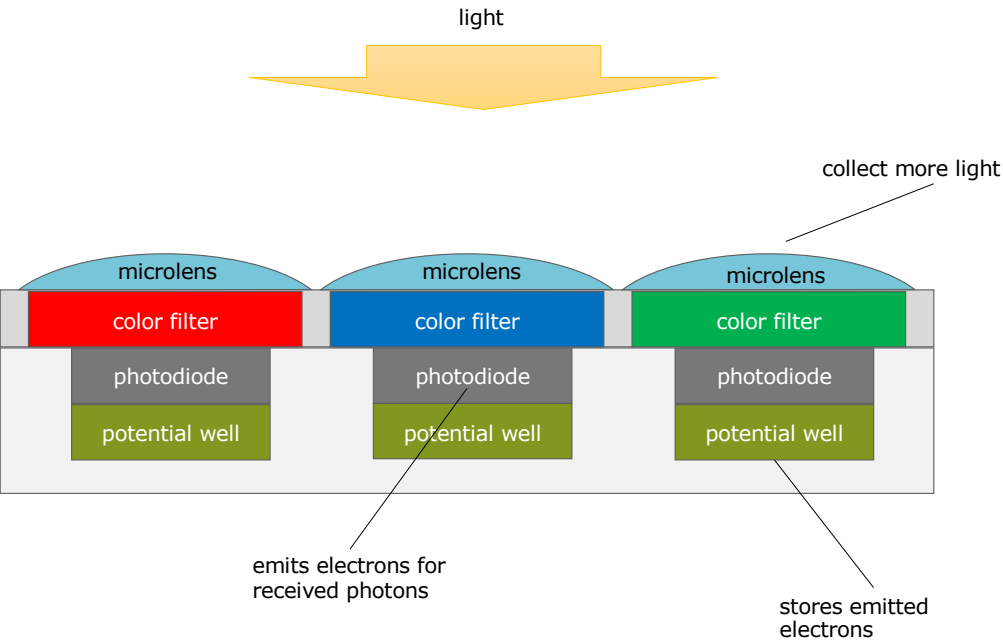
Project to measure defects in turbine blades



Photoelectronic effect



Imaging sensor



Anatomy of the Active Pixel Sensor Photodiode

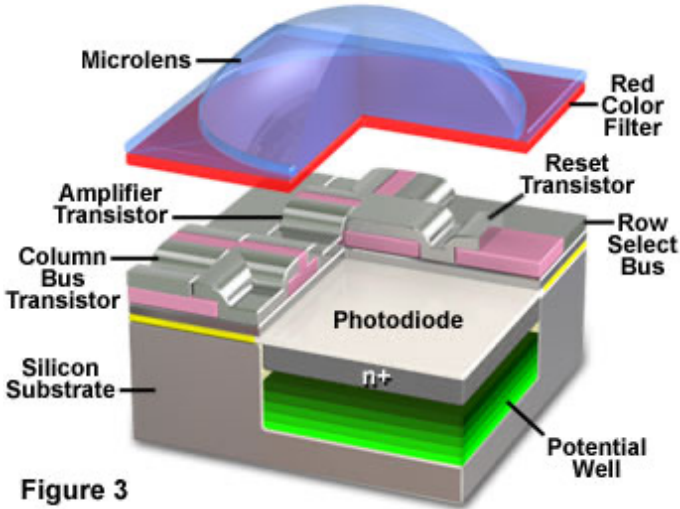
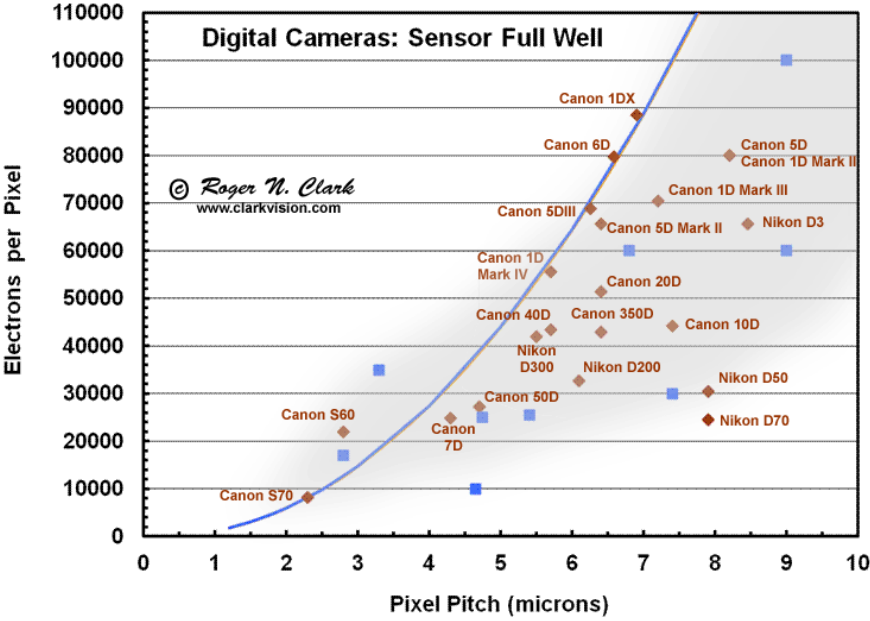


Figure 3

Photodiode capacity

How many electrons can a photodiode / well store before saturation?



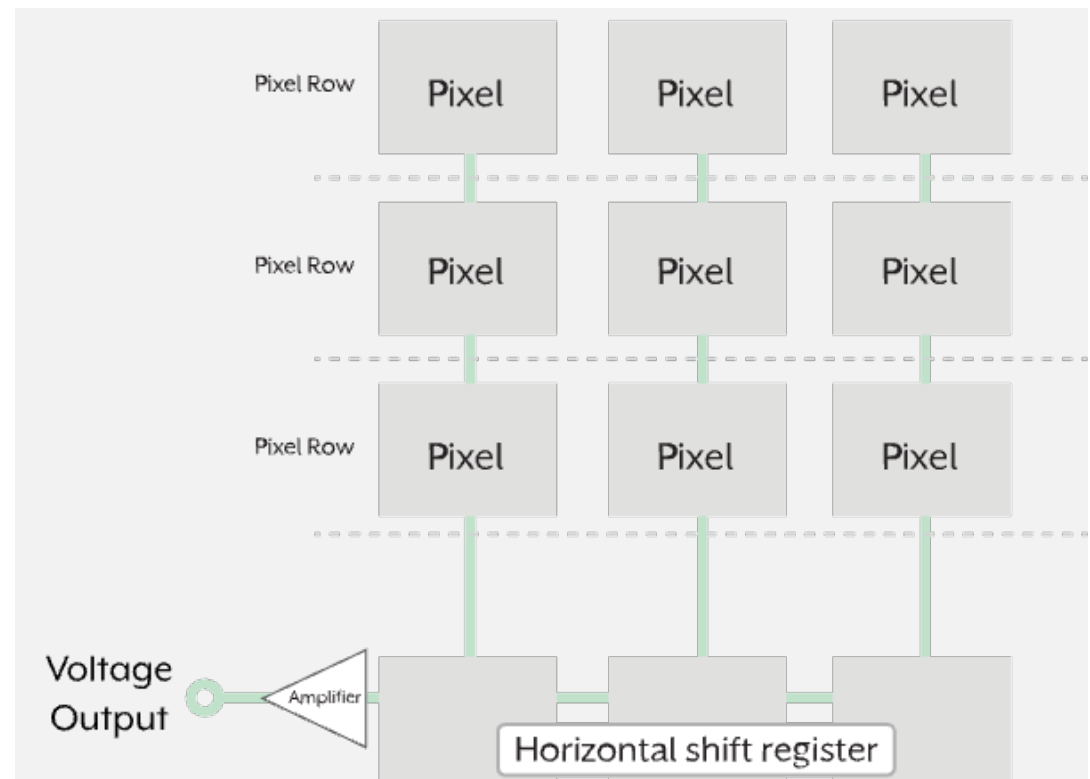
Charged Couple Device (CCD)

CCD Sensors:

- Global shutter (start / stops exposure for all pixels at the same time)
- Transfers electron charge to shift registers, then to amplifier

Characteristics:

- Low noise
- High dynamic range
- Medium range frame rates
- Subject to smearing



<https://thinklucid.com/tech-briefs/understanding-digital-image-sensors/>

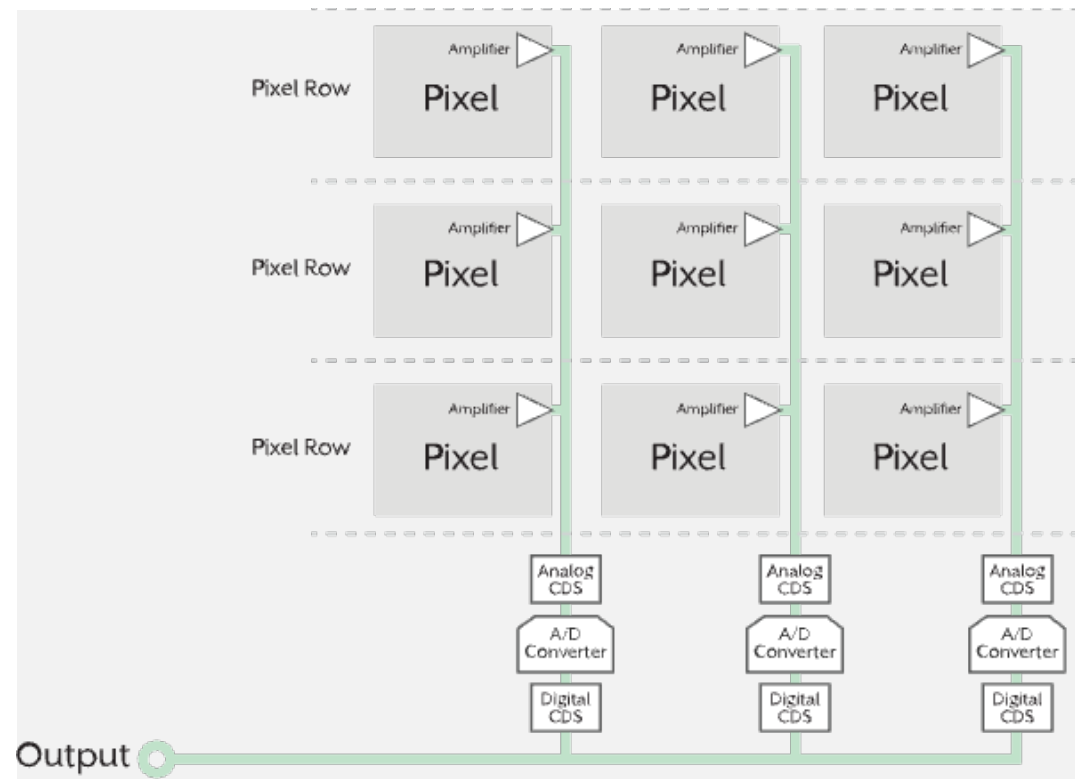
Complementary Metal-Oxide Semiconductors (CMOS)

CMOS:

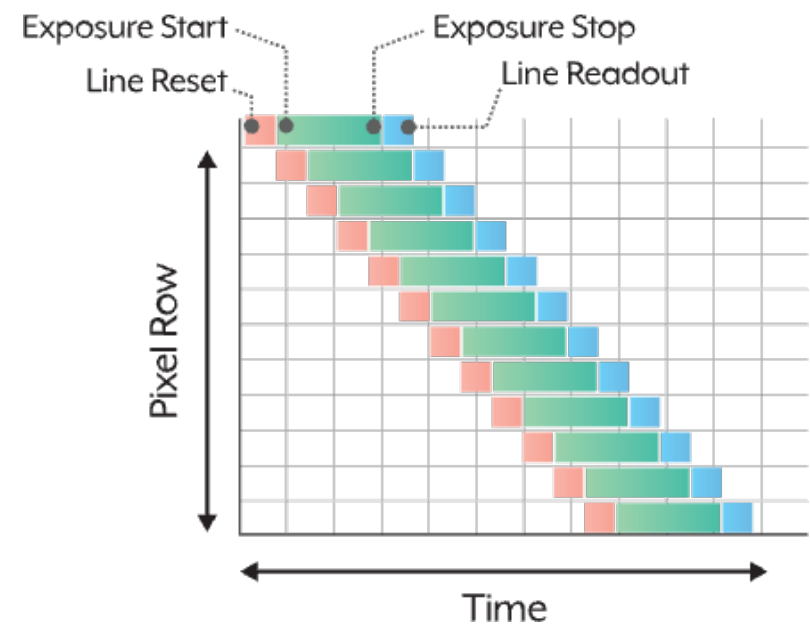
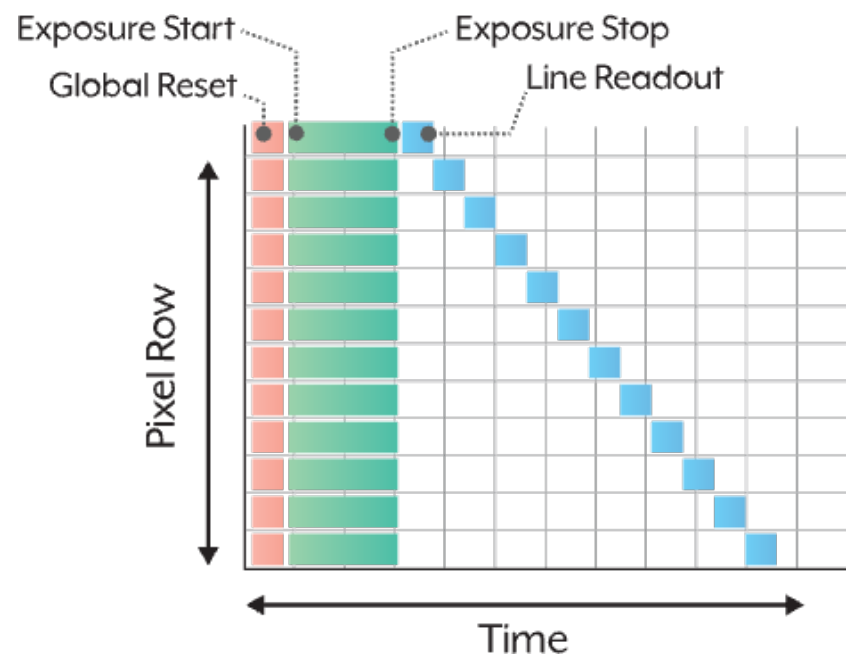
- Rolling shutter (More recent CMOS support global shutter)
- Charge converted in each pixel

Characteristics

- Low to very low noise
- Very high dynamic range
- High frame rates
- No smearing



Global Shutter Timing vs. Rolling Shutter Timing



Global Shutter Timing vs. Rolling Shutter Timing



Imaging sensor

Camera shutter opens...

- sensor converts photons into electrons at every photodiode
- stores the electrons into the potential well until full

Camera shutter closes...

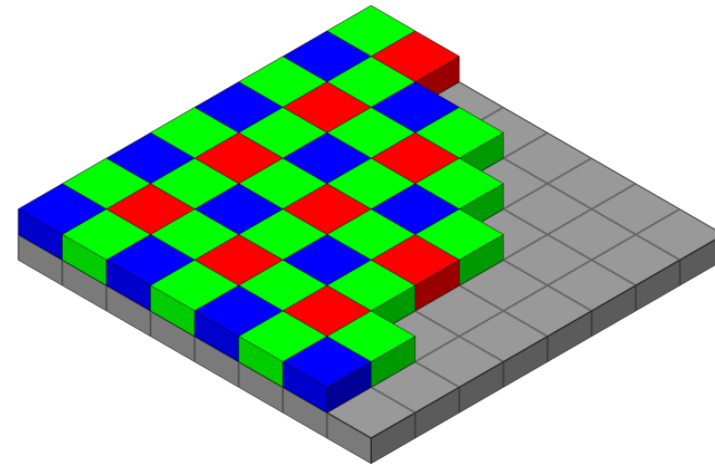
- wells are read (row by row) and converted to analog signals
- gain is applied to the signals
- analog signals are converted to digital
- nonlinearities are corrected

... an image is returned

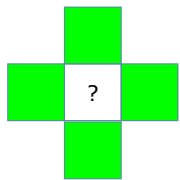
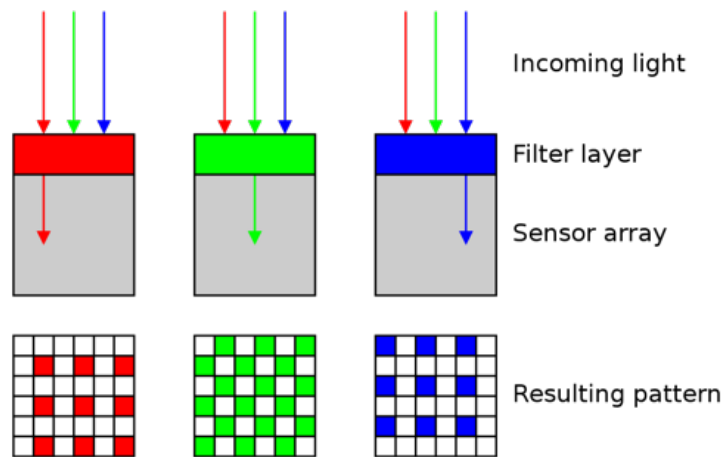
Color filter arrays (CFA)

Color sensors are arranged in a so-called color filter array (CFA)

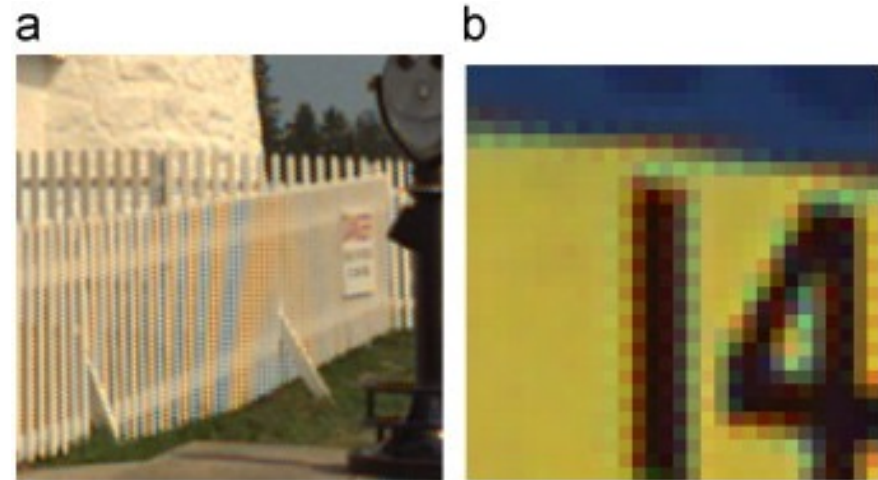
- The most common arrangement is the Bayer array
- Usually, it is done using RGB colors, but can also be used with CMY
- Are there the same number of sensors for each color channel?
- How do we get the RGB image from the sensor?



Demosaicing

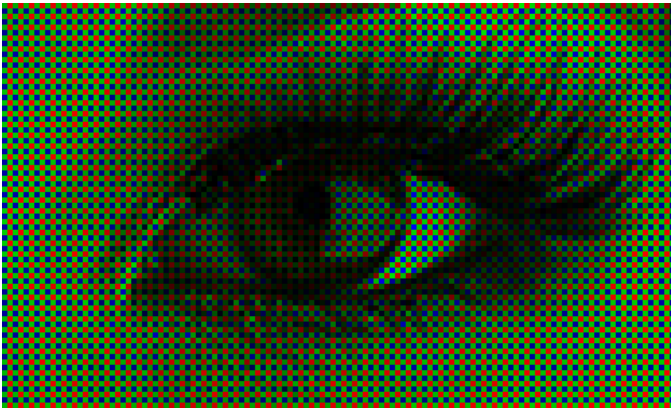


How to calculate the missing values?



Interpolation can lead to artifacts in the image.

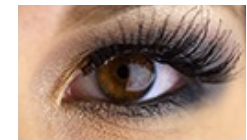
Demosaicing



Bayer image at 400%



Full color at 400%



Full color at 100%

Taking pictures

When taking pictures you have to decide on:

- Focal length
- Aperture
- Shutter speed
- ISO
- (Post processing)

and of course also

- Where to position the camera

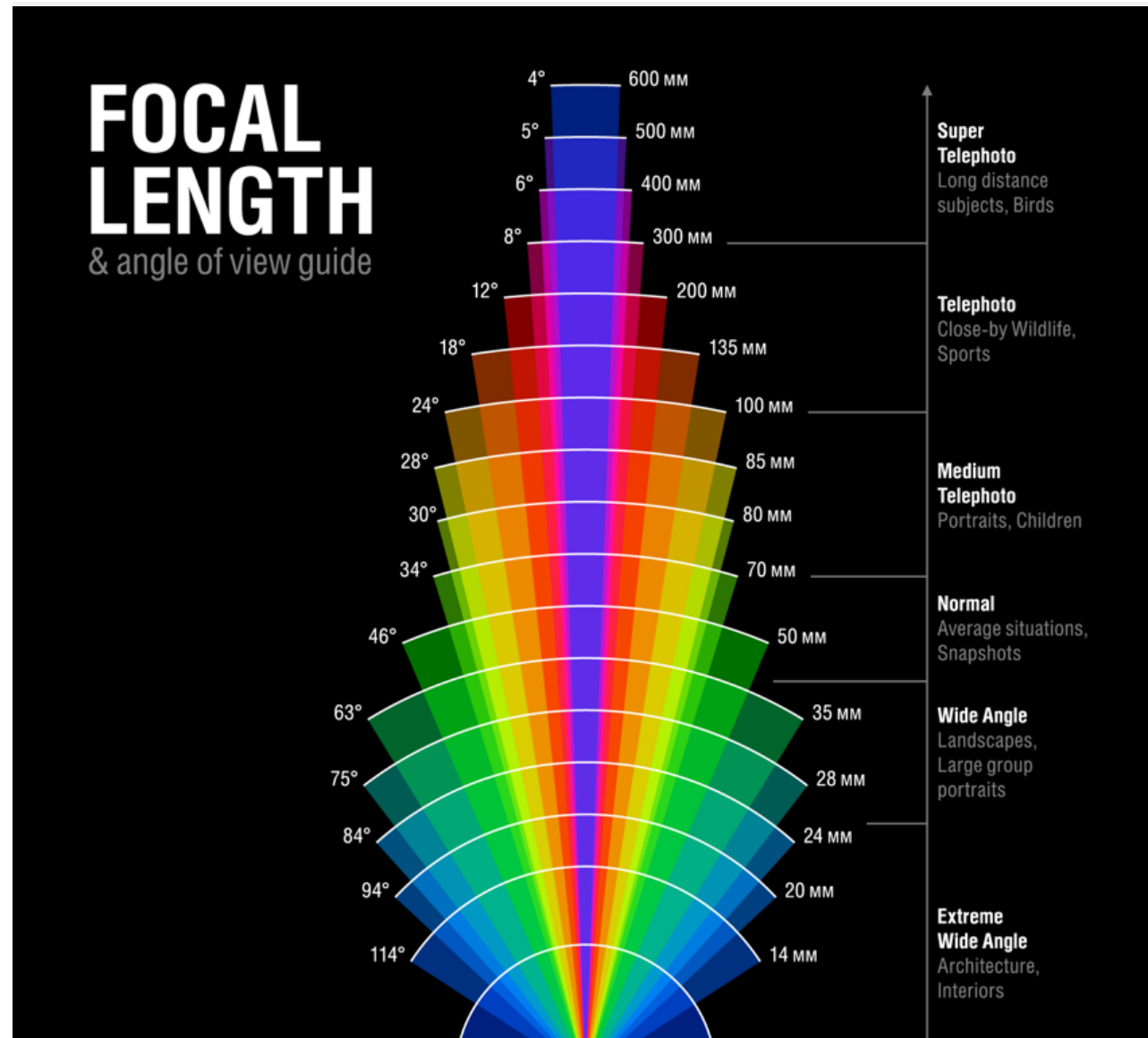
This is also important for image processing applications.

Wide angle vs. Telephoto

Focal length determines which lens to use and will influence how much of the scene you see in the image

A wide angle will allow you to capture more of the scene, but everything is smaller...

A telephoto will capture less, but in greater detail



Lenses



SLR and mirrorless cameras allow to change the lenses



Smartphones nowadays have different lenses as well

Zoom lenses provide different focal lengths

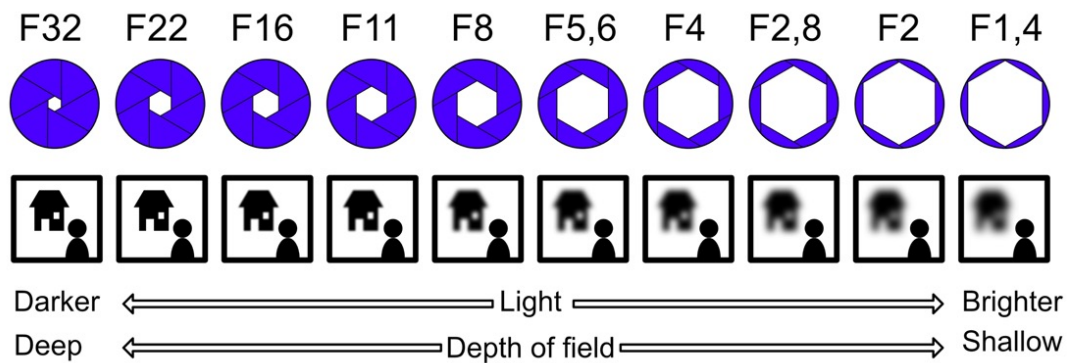
Aperture

Aperture controls the opening of the lens and is usually indicated by the *f*-number or *f*-stop:

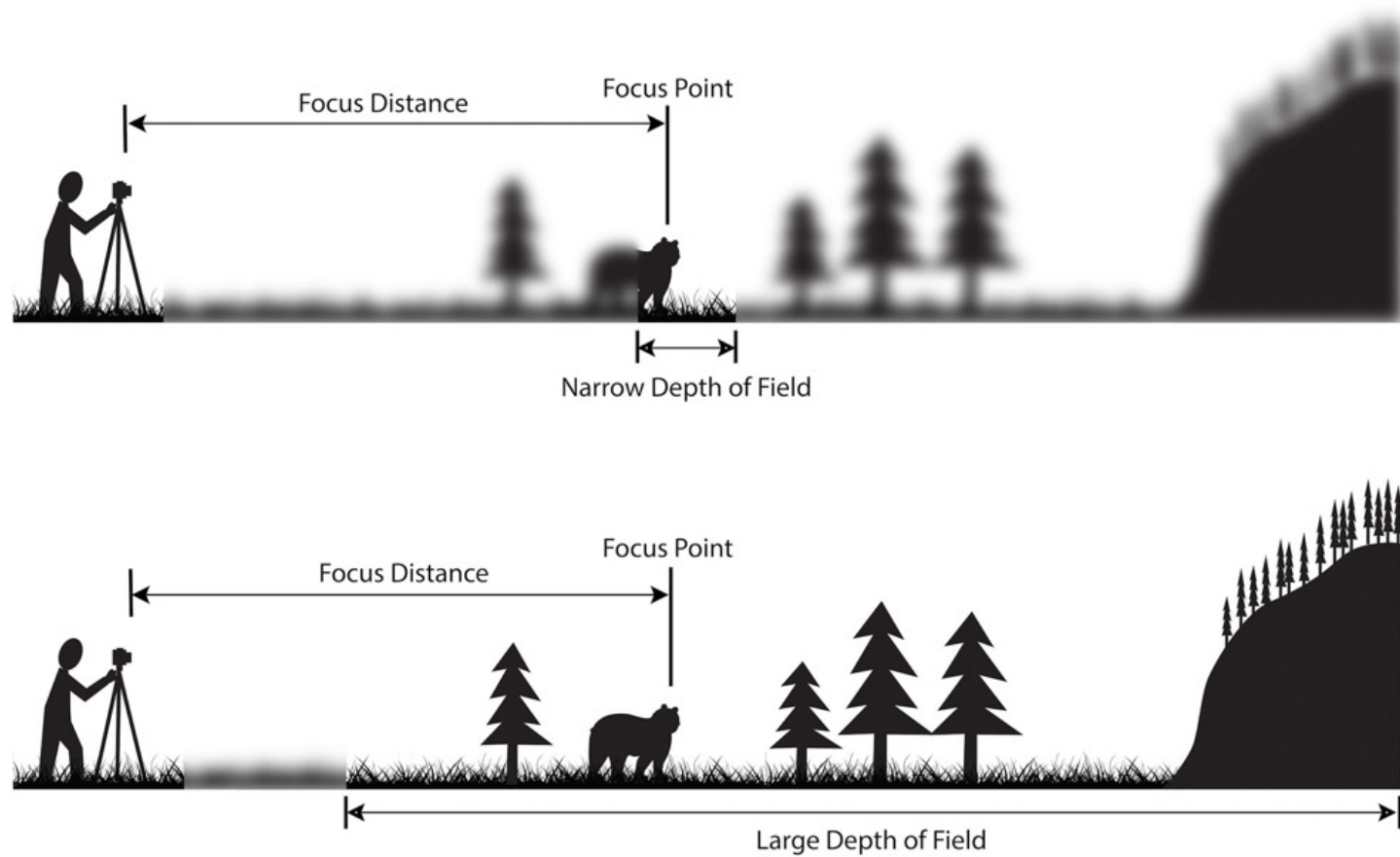
- A high *f* number corresponds to a small opening
- A small opening / high *f* number lets through less light
- A small opening / high *f* number will give a larger depth of field



Aperture



Depth of field



Depth of field



Shutter speed

Shutter speed determines how long the shutter stays open allowing light to reach the sensor.

It is typically measured in fractions of a second

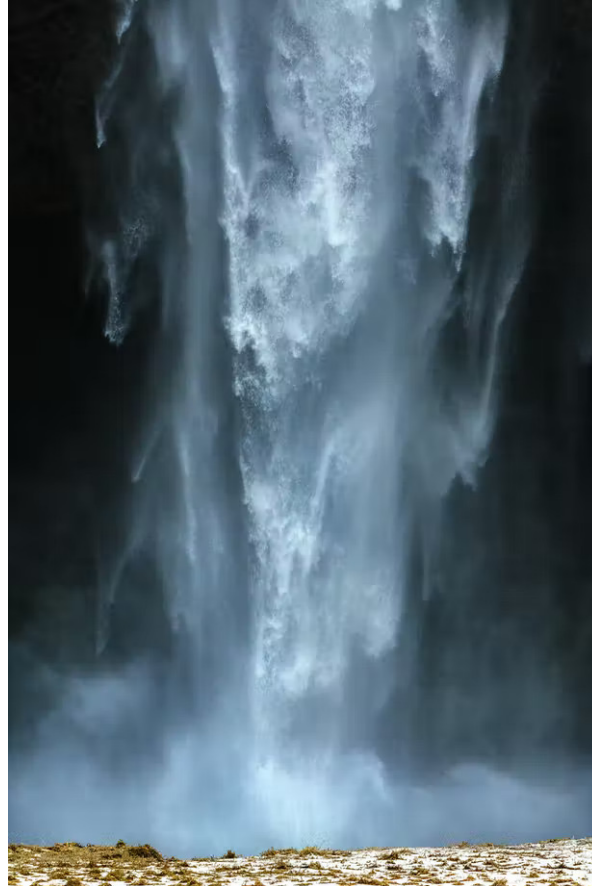
slow: $1/60$ or lower

high: $1/250$ or higher



Examples

<https://iceland-photo-tours.com/articles/photography-tutorials>



1/1000 s



30s

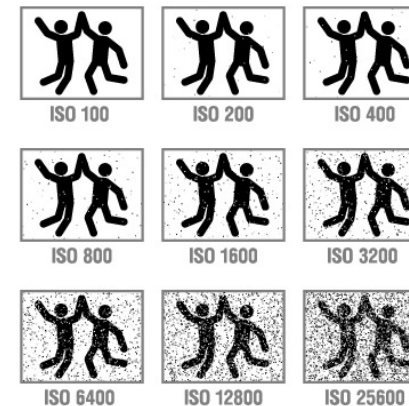
What is ISO?

In color print film (or dia film), ISO refers to the sensitivity of the film.

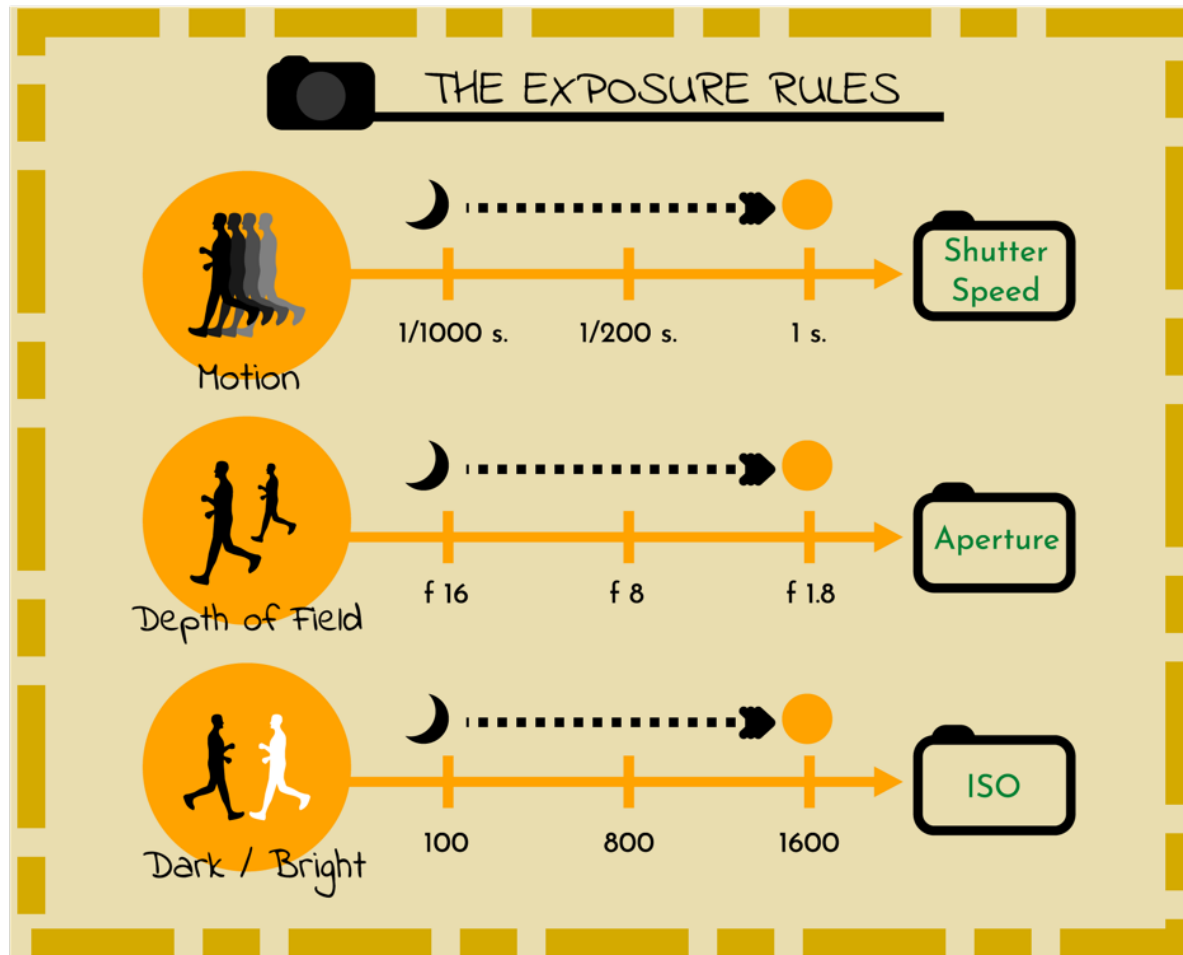
A film with higher ISO number is more sensitive and therefor needs less light.

In digital cameras, ISO will still brighten the image, by increasing the gain applied.

Higher gain will mean more noise in the image.



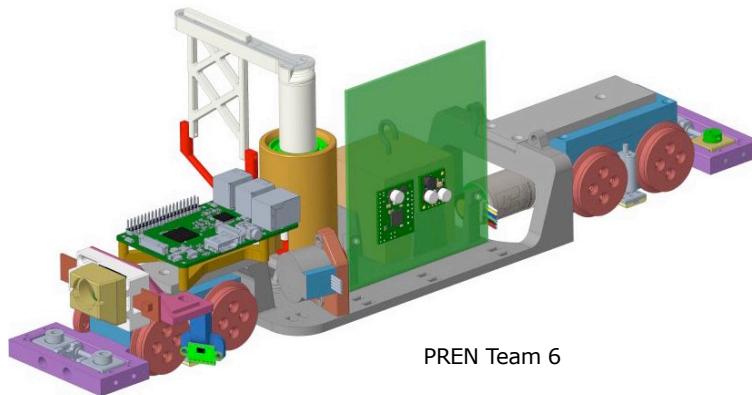
Putting it all together



Example PREN 2018/2019

Autonomous train:

- Pick up a wooden cube
- 3 rounds on track as fast as possible
- Recognize a digit on a sign near the train tracks



PREN Team 6

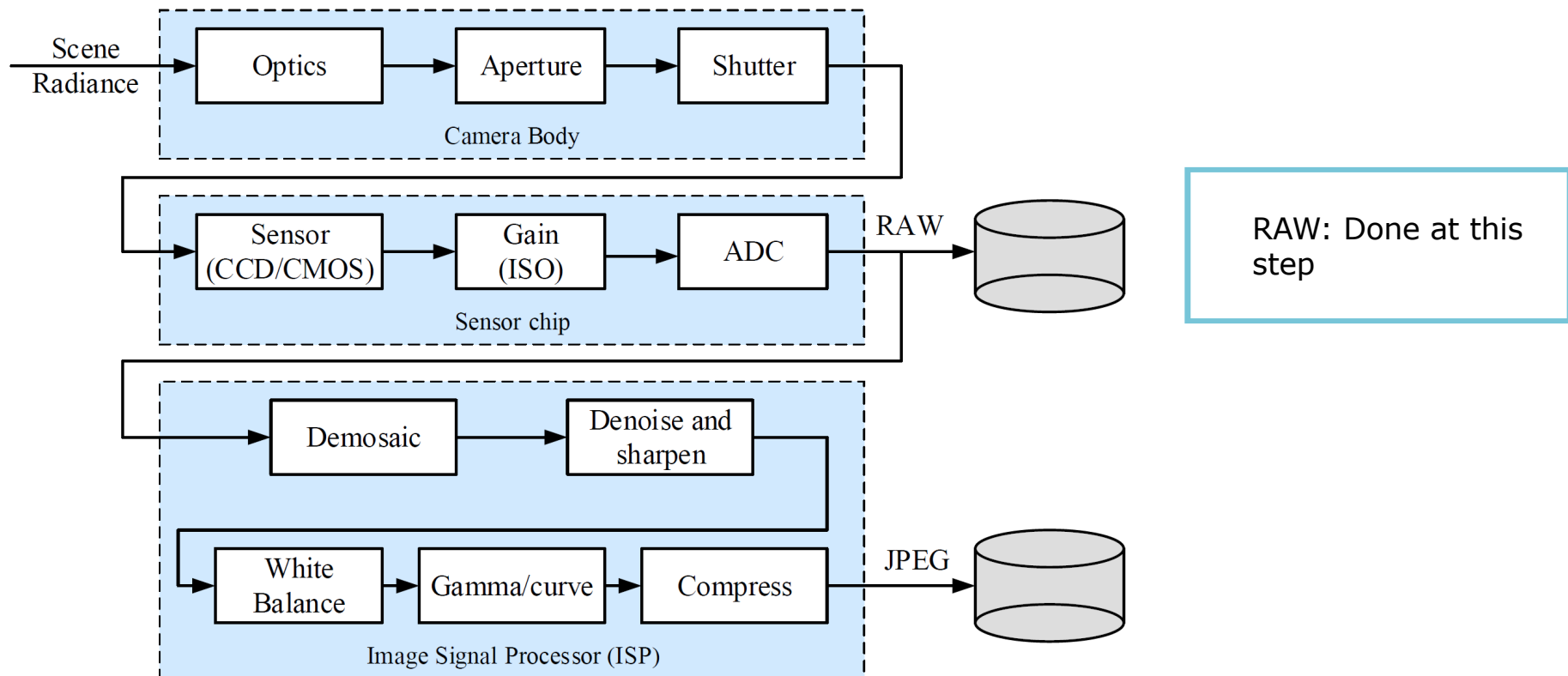


Requirements for the camera / image processing

Regarding that problem, how would you choose:

- focal length of the camera?
- aperture settings of the camera?
- shutter speed of the camera?
- ISO settings?

Digital photography pipeline



White balance

Human visual system:

- We perceive white correctly under different light sources
- Cameras do not do that (they have no perception)

Images have to be processed so that the colors that we would receive as white are rendered as white



White balance

Cameras come with large number of presets:

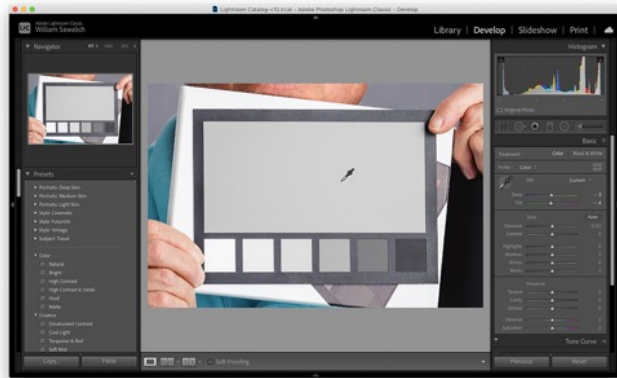
- Select under which light the image was taken, and the white balancing is applied automatically



Manual vs. automatic white balancing

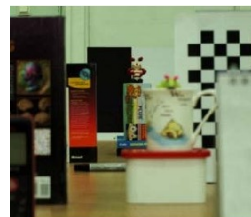
Manual white balancing:

- Select object in photograph that is color-neutral and use it to normalize
- Select camera preset based on lighting
- Postprocess white balancing

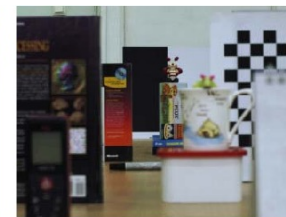


Automatic white balancing:

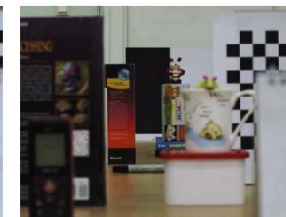
- Grey world assumption: force average color of the scene to be grey
- White world assumption: force brightest object in the scene to be white
- More sophisticated histogram-based algorithms



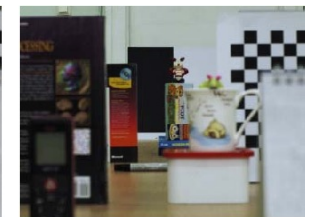
(a) Fluorescent ligh



(b) Gray world



(c) SDWGW_{1,2}



(d) YCbCr color space

Zapryanov, Georgi & Ivanova, & Nikolova, Iva. (2012). Automatic White Balance Algorithms for Digital Still Cameras - a Comparative Study. Information Technologies and Control. 1. 16-22.

Do you need to use RAW images?

In image processing, many algorithm are based on measurements of radiance

For example:

- photometric stereo
 - 3D reconstruction from multiple images
 - shape from shading
 - illumination estimation etc.
-
- I.e. these are physic-based computer vision algorithms...

I don't need physic-based vision...

You might still want to use RAW:

- Finishing your photos, for example in Adobe Lightroom
- Edits are more flexible and easier if your input is RAW

On the downside:

- Depending on the preset, you might have to *develop* all your images
- RAW will use much more memory as the files are larger, so less photos will fit your memory cards (not such a big issue nowadays)

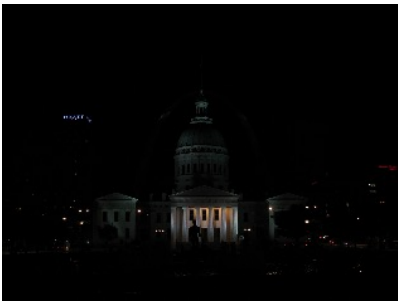
JPG vs. AppleRaw + Lightroom



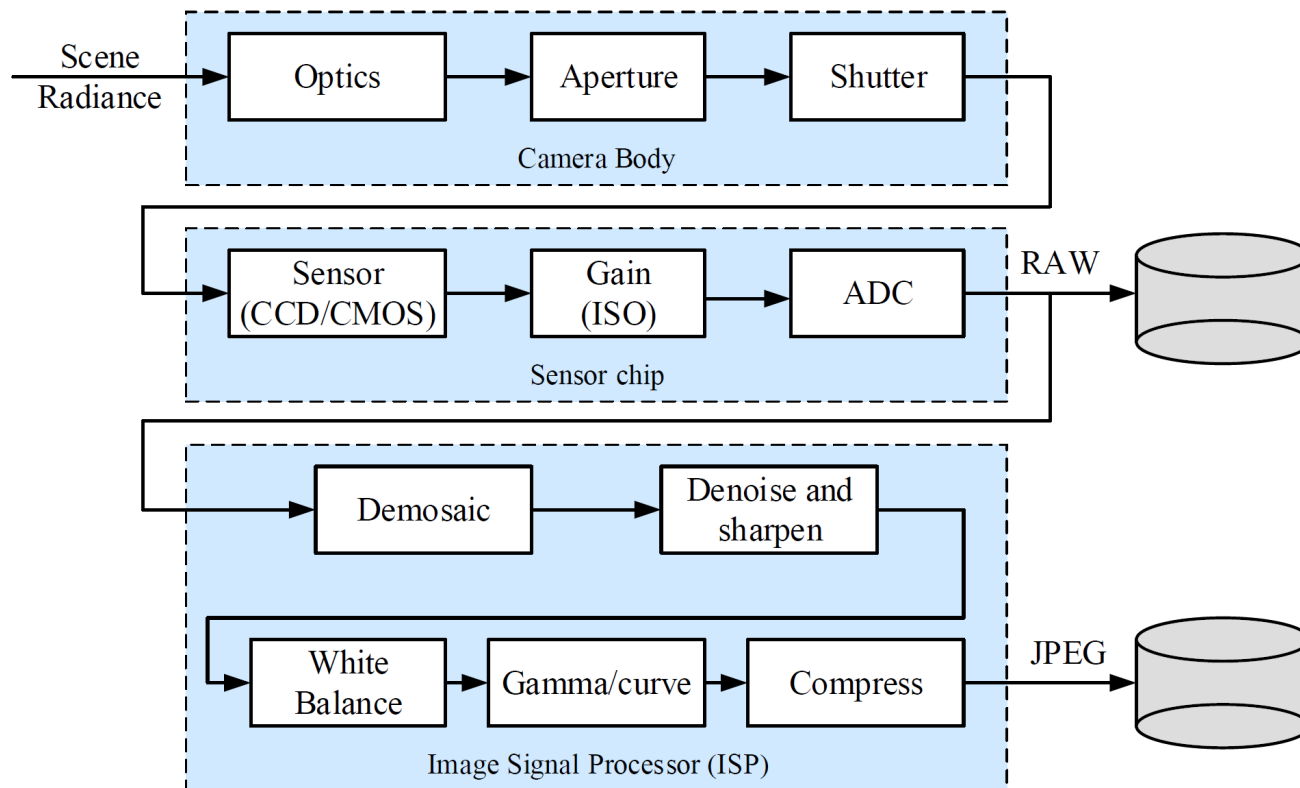
Andrew Hoyle/CNET



HDR: Creating High dynamic range images from multiple source images



Digital photography pipeline



RAW image:

- linear
- mosaiced
- 12- or 16-bit

JPEG:

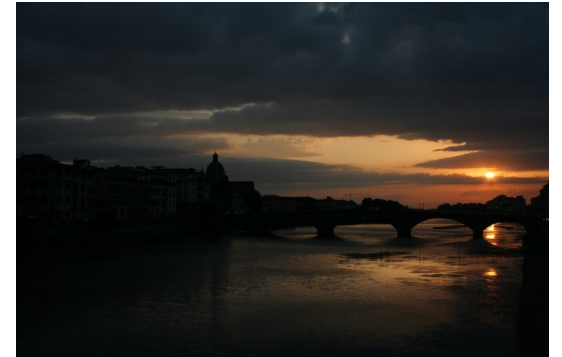
- RGB
- non-linear
- 8-bit



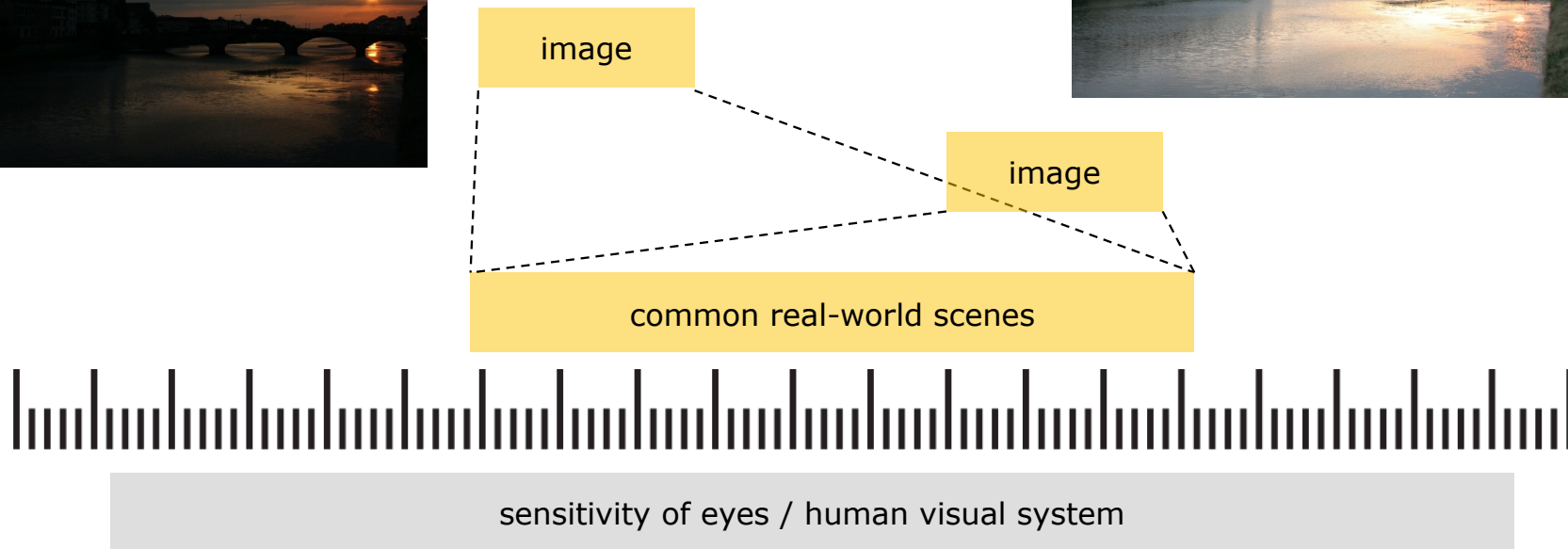
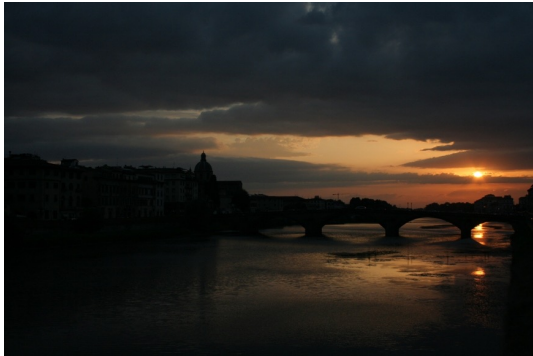
under-exposure
(noise, not enough
light)



over-exposure
(sensor saturation,
too much light)



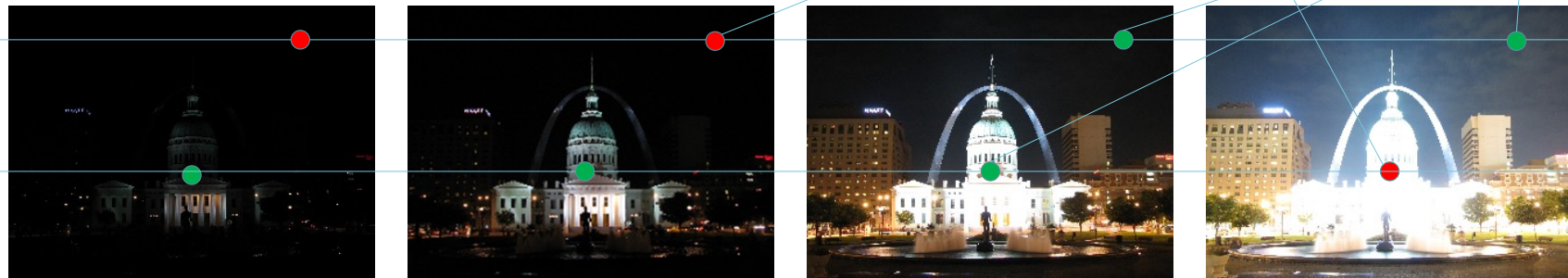
Images typically have low dynamic range



Merging RAW (linear) exposures

For each pixel:

1. Determine appropriate/valid images for this pixel
2. Weight pixel values appropriately
3. Calculate new pixel value as the weighted sum of the pixels



Exercises (nbgrader on gpuhub)

- Basic image display and operation
- Demosaicing
- Brightness corrections

- White Balance
- HDR Imaging



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